

James Watt School of Engineering

IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are entered in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers)

2 8 3 9 7 6 8
N N N N N N N

Date of Birth (DD/MM/YYYY)

0 9 / 0 3 / 2 0 0 4
D D M M Y Y Y Y

Mark with an [X] which question you have completed on this sheet.

Cross one box only.

☐ DO NOT USE

☐ DO NOT USE

☐ Q3

☐ Q4

☒ Q5

☐ Q6

☐ DO NOT USE

☐ DO NOT USE

Q5.

(a) $G_{ZAS} = (1 - z^{-1}) \cdot Z\left(\frac{G(s)}{s}\right)$ ✓

$\frac{G(s)}{s} = \frac{1}{s(s+2)(s-0.7)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s-0.7}$ ✓

$\Rightarrow A(s+2)(s-0.7) + Bs(s-0.7) + Cs(s+2) = 1$

$\Rightarrow \begin{cases} A+B+C=0 \\ 1.3A-0.7B+2C=0 \\ -1.4A=1 \end{cases} \Rightarrow \begin{cases} A = -\frac{5}{7} \approx -0.714 \\ B = \frac{5}{27} \approx 0.185 \\ C = \frac{100}{189} \approx 0.529 \end{cases}$ ✓

Therefore: $G_{ZAS} = (1 - z^{-1}) \cdot Z\left(\frac{-0.714}{s} + \frac{0.185}{s+2} + \frac{0.529}{s-0.7}\right)$
 $= (1 - z^{-1}) \cdot \left(-0.714 \cdot \frac{z}{z-1} + 0.185 \cdot \frac{z}{z-e^{-2T}} + 0.529 \cdot \frac{z}{z-e^{-0.7T}}\right)$
 $= -0.714 + 0.185 \frac{z-1}{z-0.670} + 0.529 \frac{z-1}{z-1.150}$
 $= \frac{-0.714(z-0.670)(z-1.150) + 0.185(z-1)(z-1.150) + 0.529(z-1)(z-0.670)}{(z-0.670)(z-1.150)}$
 $= \frac{0.0183z + 0.0170}{(z-0.670)(z-1.150)}$
 $= 0.0183 \cdot \frac{z + 0.929}{(z-0.670)(z-1.150)}$ ✓

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You may use both sides but only answer **ONE** question per sheet

(b) By using PI controller: $C(z) = K \frac{z-1.150}{z-1}$ ✓

TF = $C(z) \cdot G(z) = K \frac{0.0183(z+0.929)}{(z-1)(z-0.670)}$ ✓

It's ~~closed-loop~~

4/4 Its steady-state error due to a step input is 0 due to its system type number $n=1$ ✓

(c) $P_o = 100 \cdot e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}} = 15 \Rightarrow \zeta = 0.517$ ✓

The closed-loop $TF_{cl} = \frac{TF}{1+TF}$

$= \frac{K \cdot 0.0183(z+0.929)}{(z-1)(z-0.670) + K \cdot 0.0183(z+0.929)}$

$= \frac{0.0183K(z+0.929)}{z^2 + (-1.670 + 0.0183K)z + 0.670 + 0.0170K}$

$\Rightarrow \begin{cases} -2\cos(\omega_d T) \cdot e^{-\zeta \omega_n T} = -1.670 + 0.0183K \\ e^{-2\zeta \omega_n T} = 0.670 + 0.0170K \end{cases}$ ✓

$\omega_d = \omega_n \sqrt{1-\zeta^2}$

7/7 $\Rightarrow K = 2.54$ ✓

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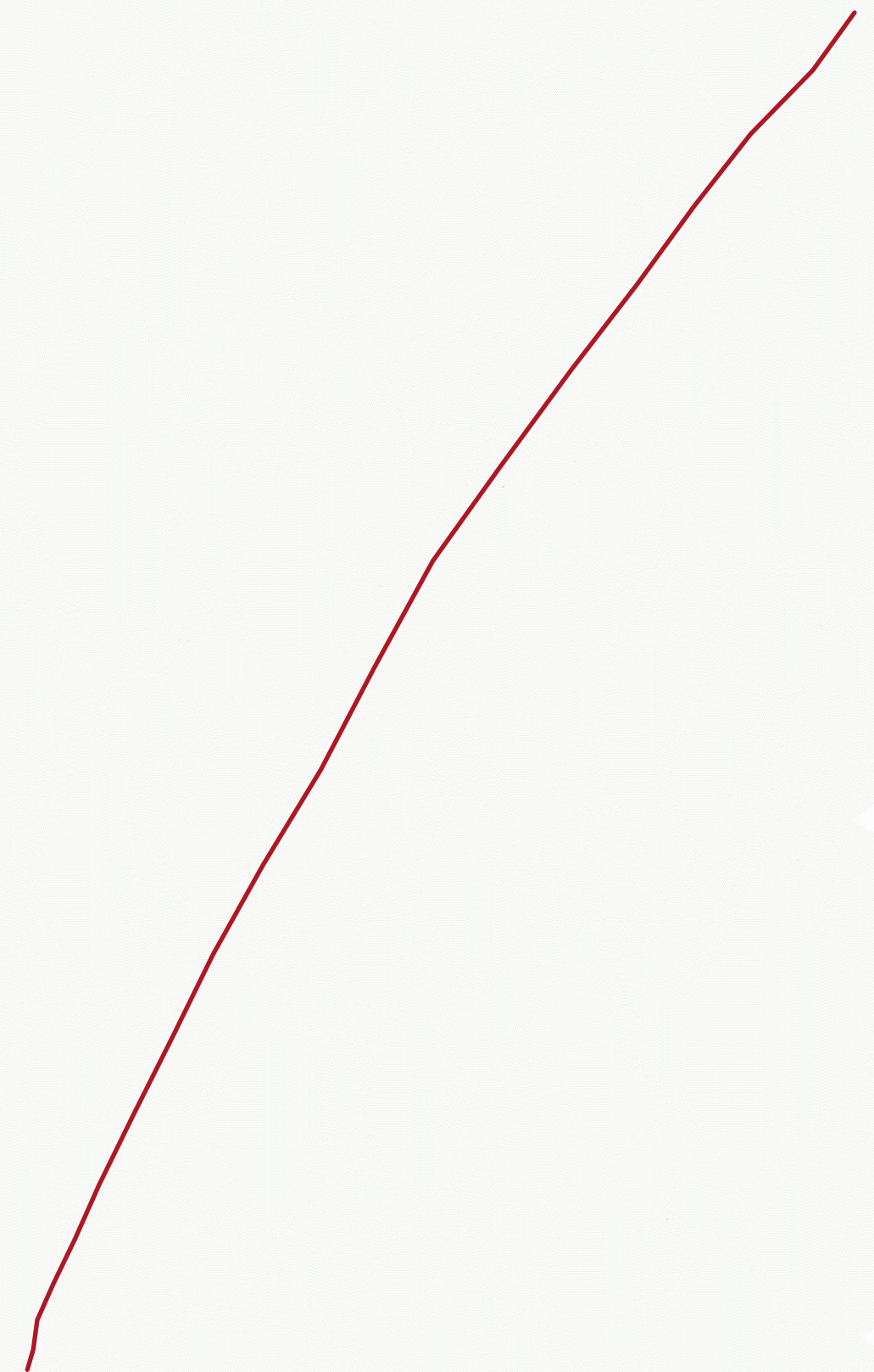
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Q Continued

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(d). When $K = 2.5$,
 $3/3$ from $e^{-2\zeta\omega_n T} = 0.670 + 0.070K$ ✓
 $\Rightarrow \omega_n = 1.639 \text{ rad/s.}$

WELL DONE!



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